

Kyle Joyce

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Education

Worcester Polytechnic Institute, Worcester, MA
Bachelor/Master's of Science in Chemical Engineering

Class of 2021/2022

Skills

Automation & Controls: Studio 5000, CompactLogix/ControlLogix, Ladder Logic, Structured Text, FactoryTalk View SE, FactoryTalk Linx, RSLinx, EtherNet/IP, OPC UA

Software & Data Systems: C#, .NET, VBA, VBScript, SQL Server, SSMS, T-SQL, XML, ODBC/ADODB, FactoryTalk Historian, OSIsoft PI, Windows Server

Bioprocess Development: Upstream and downstream processing, fermentation, bioreactor scale-up design, AKTA chromatography, tangential flow filtration, HPLC, ion-exchange chromatography, process optimization, commissioning, troubleshooting, process data analysis, and GMP documentation

Laboratory & Analytical Methods: PCR, CRISPR/Cas9, ELISA, Gel Electrophoresis, UV-Vis Spectroscopy, Cell Culture/Assays, Centrifugation, Titration, and Oligonucleotide Synthesis.

Work Experience

Software Engineer, YMC America

Sep 2024 - Aug 2026

HPLC Automation System: Led the software and controls design, implementation, testing, and commissioning of a custom HPLC system using Studio 5000, CompactLogix/ControlLogix, FactoryTalk View SE, SQL Server, and FactoryTalk Historian.

Developed sequence and equipment-module logic for pumps, valves, instruments, and process operations, with configurable parameters linked to reusable HMI schematic groups.

Translated customer requirements into PLC logic, recipes, alarms, interlocks, trends, equipment popups, and operator-interface functionality.

Coordinated with project managers, process engineers, and field service engineers through design, deployment, and on-site software updates and maintenance.

Configured system computers, software, device IP addresses, and PLC/HMI/server communications; troubleshooted EtherNet/IP, OPC UA, and FactoryTalk communication paths.

Performed functional testing of field devices, equipment modules, sequences, recipes, alarms, database connections, historian services, and HMI functionality.

Applied the same automation-development process to custom TFF and MCSGP systems, adapting control logic, recipes, and HMI functionality to each process.

Developed reusable recipe function blocks and workflow tools using C#/.NET, VBA, SQL, XML, ODBC/ADODB, ActiveX, and historian integrations.

Supported Siemens PLC/HMI and web-based HMI integration with Rockwell-based automation systems.

Manufacturing Associate - Process Development, Moderna

Nov 2023 - Sep 2024

Manufacturing Associate, Moderna

Dec 2021 - Nov 2023

Supported pilot-scale mRNA drug-substance process development across synthesis, purification, concentration, buffer exchange, and scale-up activities.

Executed development runs to evaluate process parameters, equipment settings, yield, purity, recovery, and manufacturing consistency.

Operated and troubleshooted AKTA chromatography and tangential flow filtration systems while monitoring flow, pressure, conductivity, volumes, and filtration performance.

Supported E. coli fermentation and plasmid DNA purification to generate template material for downstream RNA manufacturing.

Performed process sampling and analytical testing, including gel electrophoresis, pH, conductivity, and osmolality, to assess process performance and product quality.

Investigated process and equipment issues, collaborated with development, manufacturing, engineering, and quality teams, and maintained GMP batch records, SOPs, equipment logs, and technical documentation.